# CHE 201 Spring 2020 HW 3

Roxanna Mendoza

TOTAL POINTS

**61.5 / 100**

QUESTION 1

## 1 Question 1 **20 / 20**

- **- 5 pts** Wrong equation in convection
- **- 5 pts** Wrong calculation in convection
- **- 5 pts** Wrong equation in radiation
- **- 5 pts** Wrong calculation in radiation
- ✓ **- 0 pts correct**

QUESTION 2

## 2 Question 2 **15 / 20**

- **- 0 pts** Correct
- **- 5 pts** Wrong work of the gas
- **- 5 pts** Wrong heat transfer of the gas
- **- 5 pts** Wrong work of the piston
- ✓ **- 5 pts Wrong potential energy of the piston**

QUESTION 3

## 3 Question 3 **0 / 20**

- **- 0 pts** Correct
- ✓ **- 20 pts Please solve the revised Problem2.49 posted on the blackboard.**
- **- 5 pts** Wpaddle=-3 not 3 Btu. Paddle wheel transfers energy to the gas.
- **- 5 pts** Wrong calculation

QUESTION 4

## 4 Question 4 **17.5 / 20**

- ✓ **- 0 pts Correct**
- **- 5 pts** Wrong shaft work
- **- 5 pts** Wrong displacing atmosphere work
- **- 5 pts** Wrong heat transfer to the gas
- ✓ **- 2.5 pts Incomplete accounting of the heat transfer of energy**
- **- 5 pts** Wrong accounting of the heat transfer of energy

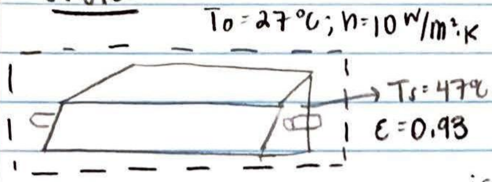
QUESTION 5

## 5 Question 5 **9 / 20**

- **- 0 pts** Correct
- ✓ **- 1 pts Wrong sketch of the cycle**
- **- 5 pts** Wrong sketch of the cycle
- ✓ **- 5 pts Wrong heat transfer for process 1-2**
- **- 5 pts** Wrong V2
- ✓ **- 5 pts Wrong Wcycle**

Homework #3 Roxanna Mendoza

#1. Sketch




Engineering Model
① The grill hood is a closed system
② For the grill hood, convection is the dominant heat transfer mode

$\dot{Q}_e = \varepsilon \sigma A [T_s^4 - T_o^4]$
↑
Stefan-Boltzmann constant ($5.67 \times 10^{-8}\ W/m^2 \cdot K^4$)

$\dot{Q}_e = 0.93 \times (5.67 \times 10^{-8}\ W/m^2 \cdot K^4)(A)[320.15^4 - 300.15^4]$

$\dot{Q}_e = 125.99 \cdot A$

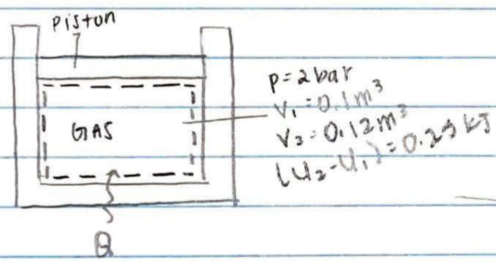
$\frac{\dot{Q}_e}{A} = 125.99\ \frac{W}{m^2} \rightarrow \boxed{0.126\ \frac{kW}{m^2} \text{ by radiation}}$

$\dot{Q}_c = hA(T_s - T_o)$

$\dot{Q}_c = (10\ W/m^2 \cdot K)\ A\ (320.15 - 300.15)$

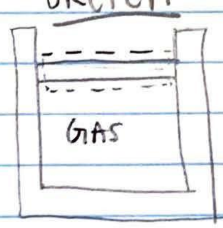
$\frac{\dot{Q}_c}{A} = 200\ \frac{W}{m^2} \rightarrow \boxed{0.2\ \frac{kW}{m^2} \text{ by convection}}$

#2.




Engineering Model
① system is a closed system
② work is only in the form of volume change
③ piston and cylinder walls are fabricated from heat resistant material

B. Sketch



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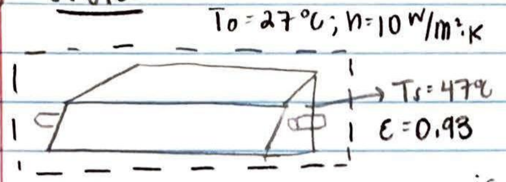


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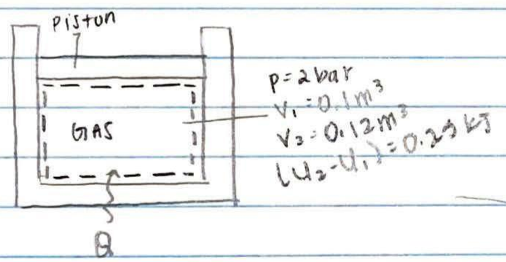


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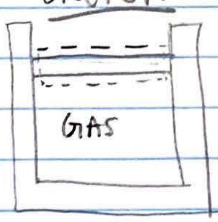

Part A:
$$W = \int_{V_1}^{V_2} P\,dV \Rightarrow W = p(V_2 - V_1)$$

$$\frac{2\text{ bar}}{} \bigg| \frac{10^5\text{ Pa}}{1\text{ bar}} = 2 \times 10^5\text{ Pa}$$

$$W = (2 \times 10^5\text{ Pa})(0.12\text{ m}^3 - 0.1\text{ m}^3)$$
$$W = 4{,}000\text{ Pa}\cdot\text{m}^3 \left(\frac{1\text{ kJ}}{10^3\text{ Pa}\cdot\text{m}^3}\right)$$
$$W = 4\text{ kJ}$$

$$Q = W + \Delta U$$
$$Q = 4\text{ kJ} + 0.25\text{ kJ}$$
$$Q = 4.25\text{ kJ}$$

Part B:
$$W = \int_{V_1}^{V_2} (P_{atm} - P_{gas})\,dV$$

$$W = (P_{atm} - P_{gas})(V_2 - V_1)$$
$$(P_{atm} - P_{gas}) \Rightarrow (1 - 2\text{ bar}) = -1\text{ bar}$$
$$\frac{-1\text{ bar}}{} \bigg| \frac{10^5\text{ Pa}}{1\text{ bar}} = -1 \times 10^5\text{ Pa}$$
$$W = (-1 \times 10^5\text{ Pa})(0.12\text{ m}^3 - 0.1\text{ m}^3)$$
$$W = -2000\text{ Pa}\cdot\text{m}^3 \left(\frac{1\text{ kJ}}{10^3\text{ Pa}\cdot\text{m}^3}\right)$$
$$W = -2\text{ kJ}$$

$$Q = W + \Delta U$$
$$Q = -2 + 0.25\text{ kJ}$$
$$Q = -2.25\text{ kJ}$$

## 2 Question 2 **15 / 20**

- **- 0 pts** Correct
- **- 5 pts** Wrong work of the gas
- **- 5 pts** Wrong heat transfer of the gas
- **- 5 pts** Wrong work of the piston
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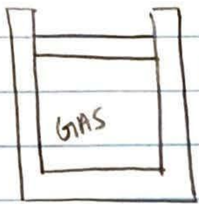



$A = 40 in^2$
$W = 100 lbf$
$P_{atm} = 14.7 lbf/in^2$
$E = 3$ Btu
$\Delta y = 1 ft = 12$ inches

Engineering malel
① ◦ closed system
② ◦ The piston and cylinder are poor thermal conductors
③ ◦ Friction can be neglected

ΔPE for the piston

$$\Delta PE = |W + (P_{atm} \times A)| \times \Delta y$$

$$\Delta PE = |(100 lbf) + (14.7 lb/in^2 \times 40 in^2)| \times (12 inches)$$

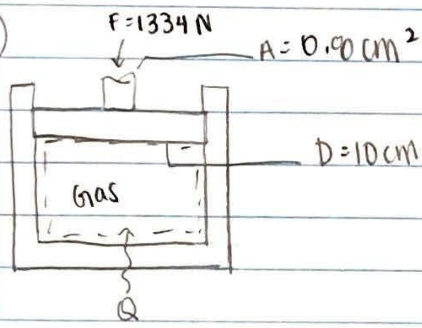
$$\Delta PE = 7156 \; lbf \cdot in$$

$$7156 \; lbf \cdot in \left| \frac{1.07 \times 10^{-4} \; Btu}{1 \; lbf \cdot in} \right. = 0.7657 \; Btu$$

$$W_{gas} = -\Delta PE = -0.7657 \; Btu$$

$$Q - W = \Delta U_{gas} \Rightarrow 0 - (-0.7657) = .7657 \; Btu$$

#4.




Engineering model
① ◦ closed system
② ◦ heat transfer to gas
③ ◦ acceleration of gravity is constant
④ ◦ piston & cylinder are poor conductors

Part A: First calculate change in elevation of the shaft

$$\Delta y = \frac{\Delta PE}{mg}$$

$$\Delta y = \frac{0.2 \; kJ}{(25 \; kg)(9.81 \; m/s^2)} \left[\frac{10^3 \; N \cdot m}{kJ}\right]\left[\frac{kg \cdot m/s^2}{N}\right] = 0.815 \; m$$

$$W_s = F \cdot \Delta y$$

$$W_s = (1334 \; N)(0.815 \; m) \left[\frac{kJ}{10^3 \; N \cdot m}\right]$$

$$W_s = 1.0872 \; kJ$$

## 3 Question 3 **0 / 20**

- **0 pts** Correct
- ✓ **- 20 pts** **Please solve the revised Problem2.49 posted on the blackboard.**
- **5 pts** Wpaddle=-3 not 3 Btu. Paddle wheel transfers energy to the gas.
- **5 pts** Wrong calculation

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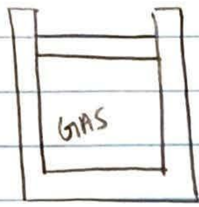


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Engineering model
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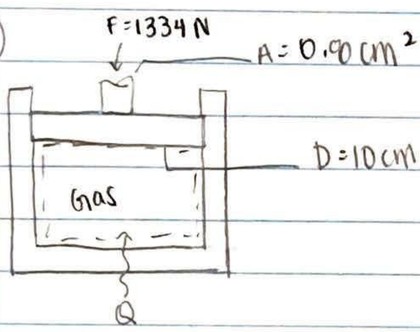


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$$W_s = F \cdot \Delta y$$

$$W_s = (1334\,N)(0.815\,m)\left[\frac{kJ}{10^3\,N \cdot m}\right]$$

$$W_s = 1.0872\,kJ$$

Part B:

Calculate the area of piston

$$A_{net} = A_{piston} - A_{shaft}$$
$$= \pi r^2 - A_{shaft}$$
$$= \pi (5cm)^2 - 0.80 cm^2$$
$$= 77.74 cm^2$$

calculating work

$$W_{atm} = (P_{atm} A_{net}) \Delta y$$

$$W_{atm} = \left[(1 \text{ bar}) \frac{10^5 N/m^2}{bar}\right]\left[(77.74 cm^2)\left[\frac{m}{100cm}\right]^2\right]\left[\frac{kJ}{10^3 N \cdot m}\right](0.815m)$$

$$W_{atm} = 0.634 \text{ kJ}$$

Part C:

$$Q_{net} = 1.0872 kJ + 0.634 kJ + 0.2 kJ + 0.1 kJ$$
$$Q_{net} = 2.02 kJ$$

Part D:

There is an increase in elevation of shaft which means there's heat being added to the gas. And that heat is equal to the work done on the shaft. There is also internal energy done by the heat which is a given in this problem as well as potential energy which is accounted for in the $Q_{net}$. Lastly, the work done by displacing the atmosphere is as a result of heat. All of those factors were accounted for and added to determine the heat transfer to the gas.

## 4 Question 4 **17.5 / 20**

✓ **- 0 pts** **Correct**

**- 5 pts** Wrong shaft work

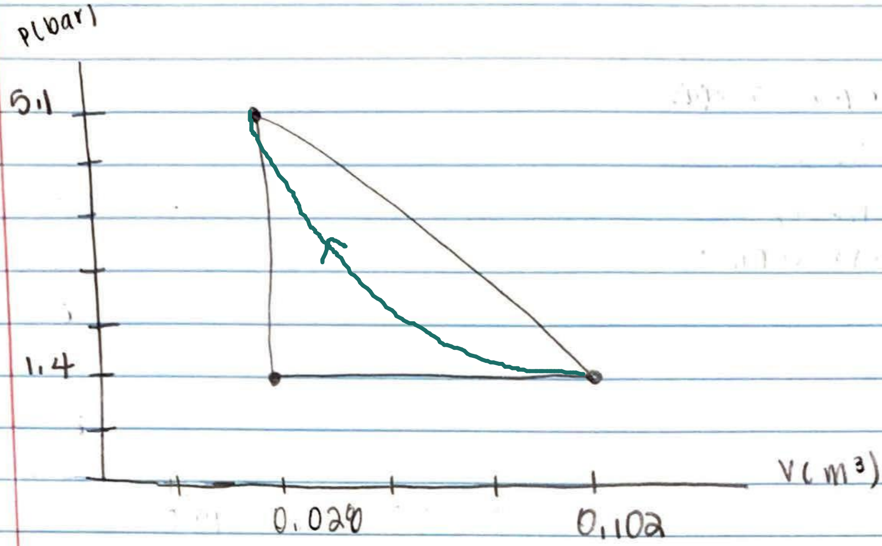
**- 5 pts** Wrong displacing atmosphere work

**- 5 pts** Wrong heat transfer to the gas

✓ **- 2.5 pts** **Incomplete accounting of the heat transfer of energy**

**- 5 pts** Wrong accounting of the heat transfer of energy

#5.




Process 1→2

- constant pressure, $p = 1.4$ bar, $V_1 = 0.028\ m^3$, $W_{12} = 10.5$ kJ

Process 2→3

- $PV$ = constant, internal energy @ Point 3 $U_3 = U_2$

Process 3→1

$V$ = constant, internal energy difference $U_1 - U_3 = -26.4$ kJ, $\Delta PE$ & $\Delta KE = 0$

*° process 1→2: $P = 1.4\text{ bar} * 101.3\times10^3 = 141.82\times10^3$ Pa

$W = \int p\,dV \Rightarrow W = \int 141.82\times10^3 dV \Rightarrow W = \int_{.028}^{V_2} 141.82\times10^3 dV \Rightarrow W = [141.82\times10^3 V]\Big|_{.028}^{V_2}$

$W_{1\to2} = 10.5\text{kJ} \therefore 10.5 = (141.82\times10^3)\cdot(V_2 - .028)\cdot(10^{-3})$

$V_2 = \frac{10.5}{141.82} + .028 = .102\ m^3$

*° Process 2→3: $C = 141.82\times10^3(.156) = 22123.92\text{ J} \Rightarrow P = 22123.92\text{J}/V$

$W = \int P\,dV \Rightarrow \int 22123.92 V^{-1} dV \Rightarrow [22123.92 \ln(V)]\Big|_{.102}^{.028}$

$W = 22123.92(\ln(.028) - \ln(.102))\times10^{-3} = -28.6$ kJ

$Q - W = \Delta U \quad Q = 0 + W \Rightarrow -28.6$ kJ

Part B

*° Process 1→3: No change in volume ∴ No work

$W_{net} = W_{1\to2} + W_{2\to3} + W_{1\to3} = 10.5\text{ kJ} - 28.6\text{ kJ} + 0 = -18.1\text{ kJ}$

$Q - W = \Delta U$

$Q_{3\to1} = U_1 - U_3 = -26.4$ kJ & $\Delta E = 0$ because cycle is ideal

$(Q_{1\to2} + Q_{2\to3} + Q_{3\to1}) - (W_{1\to2} + W_{2\to3} + W_{3\to1}) = \Delta E \Rightarrow (Q_{1\to2} - 28.6 - 26.4) - (10.5 - 28.6\ 26.4) = 0$

$Q_{1\to2} = 10.5 - 28.6 - 26.4 + 28.6 + 26.4 = 10.5$ kJ  Part C

## 5 Question 5 **9 / 20**

- **- 0 pts** Correct
- ✓ **- 1 pts Wrong sketch of the cycle**
- **- 5 pts** Wrong sketch of the cycle
- ✓ **- 5 pts Wrong heat transfer for process 1-2**
- **- 5 pts** Wrong V2
- ✓ **- 5 pts Wrong Wcycle**